

1. Administrative & Study Identification

Full Study Title: A Study to Investigate Tolerability and Efficacy of Asciminib (Oral) Versus Nilotinib (Oral) in Adult Participants (≥ 18 Years of Age) With Newly Diagnosed Philadelphia Chromosome Positive Chronic Myelogenous Leukemia in Chronic Phase (Ph+ CML-CP) **Official Regulatory Title:** A Phase IIb, Multi-center, Open-label, Randomized Study of Tolerability and Efficacy of Oral Asciminib Versus Nilotinib in Patients With Newly Diagnosed Philadelphia Chromosome Positive Chronic Myelogenous Leukemia in Chronic Phase **Short Title/Acronym:** ASC4START **Protocol Number:** CABL001J12302 **NCT Number:** NCT05456191 **Posting ID:** 476376393466319393 **Sponsor / Company Name:** Novartis / Novartis Pharmaceuticals **Investigational Product:** Asciminib (Oral) / Nilotinib (Oral) **Phase of Development:** Phase 3 (Phase IIb) **Trial Status:** Active, Not Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the tolerability of oral asciminib (80 mg QD) in comparison with the second-generation Tyrosine Kinase Inhibitor (TKI) nilotinib (300 mg BID) in adult patients with newly diagnosed Philadelphia Chromosome Positive Chronic Myelogenous Leukemia in Chronic Phase (Ph+ CML-CP), as measured by the time to study treatment discontinuation due to adverse events (TTDAE). **Secondary Objectives:** To evaluate the Major Molecular Response (MMR) and MR4.0 rates at and by scheduled data collection time points, as well as overall safety and Quality of Life (QoL) outcomes.

2.2 Background & Rationale Tyrosine kinase inhibitors (TKIs) have greatly improved treatment outcomes for patients with newly diagnosed CML-CP, transforming it into a chronic disease requiring long-term treatment. However, patients frequently experience lack of efficacy or intolerance to available treatments. Asciminib is a first-in-class BCR::ABL1 inhibitor targeting the ABL myristoyl pocket (STAMP). Its high specificity can reduce off-target effects, improving tolerability and reducing the need for therapy switching compared to second-generation TKIs like nilotinib.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Nilotinib) **Blinding:** Open-label **Randomization:**

1:1 ratio to asciminib or nilotinib (no crossover allowed). Stratified based on European Treatment Outcome Study (EUTOS) long-term survival (ELTS) score (low versus intermediate versus high). **Additional Study Phases:** Eligible participants on both arms may choose to participate in an optional 2-year Treatment Free Remission (TFR) Phase. Participants with loss of Major Molecular Response (MMR) during the TFR Phase will enter a Treatment Re-initiation (TRI) Phase.

3.2 Planned Enrollment & Duration Target **N**: 568 participants
Number of Sites: 120 sites across 24 countries **Estimated Study Start:** 12-Oct-2022 **Estimated Primary Completion:** 15-Feb-2027

4. Subject Selection (Eligibility Criteria)

4.1 Key Inclusion Criteria 1. Age ≥ 18 years with an ECOG performance status of 0 or 1. 2. Diagnosis of CML-CP (ELN 2020 criteria) within 3 months of study entry, with cytogenetic confirmation of the Philadelphia chromosome (cryptic Ph confirmed by FISH). Standard disease markers apply (e.g., $< 15\%$ blasts in blood/bone marrow, $< 20\%$ basophils, Platelet count $\geq 100 \times 10^9/L$, no extramedullary involvement except hepatosplenomegaly). 3. Evidence of typical BCR::ABL1 transcript [e14a2 and/or e13a2] amenable to standardized RQ-PCR quantification by central laboratory. 4. Adequate end organ function: Total bilirubin (TBL) $< 3 \times$ ULN; Creatinine Clearance (CrCl) ≥ 30 mL/min; Serum lipase $\leq 1.5 \times$ ULN. 5. Laboratory values within normal limits (or corrected to normal with supplements): Potassium, Total calcium (corrected for serum albumin), and Magnesium. 6. *Optional TFR Phase Inclusion:* Minimum of 3 years (up to max 5 years) of treatment on study; sustained MR4.5 (BCR:ABL1 $\leq 0.0032\%$ IS) for at least 1 year prior to entry into TFR Phase.

4.2 Key Exclusion Criteria 1. Previous treatment of CML with any other anticancer agents (including chemotherapy/biologic agents) or prior stem cell transplant, except hydroxyurea and/or anagrelide. 2. Known cytopathologically confirmed central nervous system (CNS) infiltration. 3. Impaired cardiac function including history of MI, angina pectoris, or CABG within 6 months; clinically significant arrhythmias, complete left bundle branch block, high-grade AV block; QTcF ≥ 450 ms; Long QT syndrome or risk factors for Torsades de Pointes (TdP). 4. Severe/uncontrolled concurrent medical disease (e.g., uncontrolled diabetes, active infection, uncontrolled hypertension or hyperlipidemia). 5. History of significant congenital/acquired bleeding disorder; major surgery within 4 weeks; history of other active malignancy within 3 years. 6. History of acute pancreatitis within 1 year or chronic pancreatitis. 7. Chronic/acute liver disease; known history of HBV, HCV, or HIV (unless well-controlled on stable ART). 8. Impairment of gastrointestinal (GI) function that may alter absorption (e.g., malabsorption,

gastric bypass). 9. Pregnant or nursing women. Women of child-bearing potential must use highly effective contraception. 10. *TFR & TRI Phase Exclusions:* Participants in the TRI Phase cannot re-enter the TFR Phase. Pregnancy during TFR requires discontinuation upon loss of MMR and precludes entry into the TRI phase.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Asciminib 80 mg once daily (QD) OR Nilotinib (Trade Name: Tasigna; ATC Code: L01EA03) 300 mg twice daily (BID). (*Note: Refer to EMA prescribing information for nilotinib/Tasigna for full product details*). **Route of Administration:** Oral, under fasting conditions. **Duration of Treatment:** Event-driven, with expected individual patient treatment duration of approximately 2 to 4.5 years (until $\sim$ 65 discontinuations due to AE are met). Includes optional 2-year Treatment Free Remission (TFR) Phase, and a Treatment Re-initiation (TRI) Phase if applicable.

5.2 Concomitant & Prohibited Therapy Permitted: Hydroxyurea and/or anagrelide (prior to or during early phases per protocol), and standard supportive care medications. **Prohibited:** Other antineoplastic therapies, biologic agents, or investigational agents. Medications with a "known risk of Torsades de Pointes" that cannot be safely discontinued/replaced.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Cytogenetic and PCR evaluations, ECG, Physical Exam, Viral Hepatitis/HIV screening
Baseline	Day 0	Randomization, First Dose, QoL Questionnaires
Interim	Approx. Q4W to Q12W	Blood tests, RQ-PCR for BCR::ABL1, Safety/AE Assessments, QoL Questionnaires
End of Study	After ~2 to 5 Yrs	Final Efficacy Assessment, TFR/TRI transitions if eligible, Exit Interview, IP Accountability, Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to discontinuation of study treatment due to adverse event (TTDAE). **Measurement Method:** Defined as the time from the date of first dose of study treatment to the date of discontinuation of study treatment due to an adverse event (AE).

7.2 Secondary Endpoints 1. Percentage of participants with Major Molecular response (MMR) at scheduled data collection time points. 2. Percentage of participants with Major Molecular

response (MMR) by scheduled data collection time points. 3. Percentage of participants with MR4.0 at scheduled data collection time points. 4. Additional measures include change from baseline in EORTC QLQ-C30 and QLQ-CML24 scores, and overall incidence/severity of AEs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular hospital visits, clinical assessments, blood tests, and optional bone marrow aspirations if indicated. **Serious Adverse Events (SAEs):** Routine reporting to sponsor and regulatory bodies within standard pharmacovigilance timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to review safety and efficacy data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary and secondary efficacy evaluations; Safety Set for all tolerability and AE analyses. **Sample Size Justification:** $N=568$ appropriately powered to perform the primary analysis when approximately 65 study treatment discontinuations due to an Adverse Event (AE) occur across both treatment arms. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier) for time-to-event data (TTDAE); appropriate statistical imputation for QoL measures.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. Protocol includes active management of COVID-19 related considerations. **Monitoring Plan:** Regular onsite and remote monitoring visits per sponsor Standard Operating Procedures (SOPs). **Audit Trail:** Centralized data collection via eCRFs, stringent source data verification, and RQ-PCR quantification handled by an accredited central laboratory.

11. Study Identifiers & Governance

Primary ID: CABL001J12302 **Registry Number:** NCT05456191 **Posting ID:** 476376393466319393 **Sponsor/Lead Organization:** Novartis **Study Title:** ASC4START **Official Regulatory Title:** A Phase IIb, Multi-center, Open-label, Randomized Study of Tolerability and Efficacy of Oral Asciminib Versus Nilotinib in Patients With Newly Diagnosed Philadelphia Chromosome Positive Chronic Myelogenous Leukemia in Chronic Phase

12. Clinical Framework (CML Specific)

Disease Area: Philadelphia Chromosome-Positive Chronic Myeloid Leukemia **Preferred UMLS Name:** Philadelphia Chromosome Positive Chronic Myeloid Leukemia **Clinical Codes:** MeSH: D015464 | HPO: HP: 0005506 | SNOMED: 92818009 **Primary Condition:** Philadelphia Chromosome-Positive Chronic Myelogenous Leukemia in Chronic Phase (Ph+ CML-CP) **Diagnosis Threshold:** Cytogenetic confirmation of the Philadelphia chromosome, diagnosed within 3 months **Medical Methodology:** Targeted Tyrosine Kinase Inhibitor (TKI) Therapy **Optimization Target:** Major Molecular Response (MMR) and prolonged TTDAE, leading to potential Treatment Free Remission (TFR).

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for newly diagnosed CML-CP **Education Protocol (HCP):** TKI administration guidelines, stringent AE tracking, and treatment discontinuation protocols **Education Protocol (Patient):** Fasting conditions requirements for oral dosing, strict contraception guidelines, and AE reporting self-management **Quality Audit Mechanism:** Central laboratory standardized RQ-PCR quantification and ongoing clinical assessments

14. Observation & Follow-Up Schedule

Total Duration: Approximately 2 to 4.5 years (up to 5 years for those entering TFR/TRI) **Onsite Visit Cadence:** Up to approximately 23 visits over the core treatment period **Remote Monitoring Frequency:** As per specific site/sponsor protocol and COVID-19 related guidelines **Checklist Review Interval:** Routine molecular response (MMR, MR4.0, MR4.5) tracking including evaluation points for potential TFR entry.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				